L7: Eigen-decompositions and Principal Component Analysis

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Linear Algebra in Machine Learning
Yun Kuen Cheung, School of Computing, ANU

An MNIST image has 784 pixel values, but these features do not vary independently: nearby pixels often change together as the shape or position of a handwritten digit changes. More generally, a dataset with many features may admit a much shorter description. Lecture 4 made this precise: when datapoints lie exactly in a low-dimensional subspace, each datapoint admits a shorter *coordinate vector*. Real data, however, usually lie only approximately in such a subspace. Lecture 5 showed how to project onto a given subspace and obtain its closest-point approximations.

The remaining question is how to choose that subspace from the data. *Principal Component Analysis* (PCA) chooses directions that retain as much of the dataset's variation as possible. We will show that this also minimizes the average squared reconstruction error. To derive these results, we first develop eigenvectors and the special structure of real symmetric matrices. The argument uses finite averages, dot products and orthogonal projection.

Why reduce the number of features?

A data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ records n datapoints, each with d features. Here n is the number of datapoints/samples, whereas d is the *ambient dimension*: the dimension of the space in which each datapoint is represented. Increasing n gives us more observations in the same space; increasing d gives each observation more features. More features can be useful, but they also increase computational costs and make it harder to obtain data that cover the space well.

Example 1. Suppose each of d features can take either the value 0 or 1, with all combinations allowed. There are 2^d possible feature vectors. When $d = 20$, $2^{20} = 1048576$. A dataset with $n = 1000$ samples covers at most 1000 distinct combinations, which is less than 0.1% of all possible combinations.

This example illustrates one aspect of the curse of dimensionality: covering an *unstructured* feature space can require a number of samples that grows exponentially with its dimension. This is not a claim that every learning problem requires 2^d samples. A model need not observe every possible combination if the data *have additional structure*. In particular, if the datapoints lie near a much smaller subspace, we can seek a representation with $k < d$ coordinates that preserves their important variation.

Another motivation is noise, such as measurement errors or unwanted fluctuations in the recorded features. If the useful variation lies in a subspace, projection onto that subspace removes any component of the noise perpendicular to it.

Example 2. Suppose the useful signal lies on the line $\mathcal{S} = \{t\mathbf{q} : t \in \mathbb{R}\}$, where $\mathbf{q} = \frac{1}{5}(3, 4, 0)^\mathsf{T}$ is a unit vector. A clean datapoint $\mathbf{s} = (15, 20, 0)^\mathsf{T}$ is recorded with an error $\mathbf{e} = (0, 0, 3)^\mathsf{T}$, giving the observation $\mathbf{x} = \mathbf{s} + \mathbf{e} = (15, 20, 3)^\mathsf{T}$. Since $\mathbf{e}$ is perpendicular to $\mathbf{q}$, projection recovers the clean datapoint:

$$\langle \mathbf{q}, \mathbf{x} \rangle \, \mathbf{q} \;=\; 25\mathbf{q} \;=\; \mathbf{s}.$$

In practice, we do not know the useful subspace in advance. PCA chooses a subspace from the observed variation, rather than from knowledge of the clean signal. Discarding low-variance directions can reduce

noise when those directions mainly contain noise, but *discarded variation is not automatically noise*: a low-variance direction may still contain information important for a prediction task.

Eigenvectors identify directions a matrix preserves

A general matrix mixes the coordinates of an input vector. If we could instead find a basis in which each basis vector is merely scaled, the transformation would be easier to understand: each coordinate would be multiplied by its own scalar. PCA will use such directions for a matrix that describes variation in data.

Definition 3. Let $\mathbf{A} \in \mathbb{R}^{d\times d}$. A nonzero vector $\mathbf{v} \in \mathbb{R}^d$ is an eigenvector of $\mathbf{A}$ with eigenvalue $\lambda \in \mathbb{R}$ if

$$\mathbf{A}\mathbf{v} \;=\; \lambda\mathbf{v}.$$

For an eigenvalue λ, its eigenspace is

$$\mathcal{E}_\lambda \;=\; \{\mathbf{v} \in \mathbb{R}^d : \mathbf{A}\mathbf{v} = \lambda\mathbf{v}\} \;=\; \mathrm{Nul}(\mathbf{A} - \lambda\mathbf{I}_d).$$

For $\lambda > 0$, the map scales an eigenvector without reversing its direction. For $\lambda < 0$, it also reverses the direction. For $\lambda = 0$, it maps the eigenvector to zero. We exclude the zero vector from the definition because $\mathbf{A0} = \lambda\mathbf{0}$ holds for every scalar λ. The eigenspace nevertheless includes the zero vector. An eigenspace is a subspace, as its null-space description shows. Consequently, any nonzero multiple of an eigenvector is another eigenvector with the same eigenvalue. Once an eigenvalue is known, Gaussian elimination on $(\mathbf{A} - \lambda\mathbf{I}_d)\mathbf{v} = \mathbf{0}$ finds its eigenspace.

Example 4. Let

$$\mathbf{A} \;=\; \begin{pmatrix} 1 & 10 & -3 \\ 10 & 4 & 18 \\ -3 & 18 & -15 \end{pmatrix}, \qquad \mathbf{v}_1 \;=\; \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}, \qquad \mathbf{v}_2 \;=\; \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \qquad \mathbf{v}_3 \;=\; \begin{pmatrix} -4 \\ 1 \\ 2 \end{pmatrix}.$$

The three vectors are eigenvectors of $\mathbf{A}$ with eigenvalues -28, 18 and 0, respectively:

$$\mathbf{A}\mathbf{v}_1 \;=\; -28\mathbf{v}_1, \qquad \mathbf{A}\mathbf{v}_2 \;=\; 18\mathbf{v}_2, \qquad \mathbf{A}\mathbf{v}_3 \;=\; \mathbf{0}.$$

In general, computing the eigenvalues of a matrix by hand is not easy, especially when its dimension is large. Calculations for 2×2 and 3×3 matrices remain feasible but can still require substantial algebra.

Not every real square matrix has a basis of real eigenvectors. For PCA, we will work with a class of matrices for which an even stronger statement holds: an orthonormal eigenbasis always exists.

Symmetric matrices have orthonormal eigenbases

Coordinates relative to an arbitrary basis need not preserve lengths and angles. For a method based on distances and projection, we would prefer an orthonormal basis, whose coordinates preserve this geometry. Symmetric matrices provide exactly the eigenbases we need.

Definition 5. A matrix $\mathbf{A} \in \mathbb{R}^{d\times d}$ is symmetric if $\mathbf{A}^\mathsf{T} = \mathbf{A}$, or equivalently, if $a_{ij} = a_{ji}$ for every i, j.

Proposition 6. Eigenvectors of a real symmetric matrix corresponding to distinct eigenvalues are orthogonal.

Proof: Suppose $\mathbf{Au} = \lambda\mathbf{u}$ and $\mathbf{Av} = \mu\mathbf{v}$, where $\lambda \neq \mu$. Symmetry gives

$$\lambda \langle \mathbf{u}, \mathbf{v} \rangle \;=\; \langle \mathbf{Au}, \mathbf{v} \rangle \;=\; \mathbf{u}^\mathsf{T}\mathbf{A}^\mathsf{T}\mathbf{v} \;=\; \mathbf{u}^\mathsf{T}\mathbf{Av} \;=\; \langle \mathbf{u}, \mathbf{Av} \rangle \;=\; \mu \langle \mathbf{u}, \mathbf{v} \rangle .$$

Thus $(\lambda - \mu)\langle \mathbf{u}, \mathbf{v} \rangle = 0$. Since $\lambda - \mu \neq 0$, the dot product is zero. □

This proposition explains orthogonality when the eigenvalues are distinct. The next theorem also guarantees that enough real eigenvectors exist, including when eigenvalues repeat.

Theorem 7. Spectral theorem for real symmetric matrices. Every real symmetric matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$ has an orthonormal basis $\mathbf{q}_1, \ldots, \mathbf{q}_d$ of eigenvectors with real eigenvalues $\lambda_1, \ldots, \lambda_d$. Define

$$\mathbf{Q} \;=\; \begin{pmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_d \\ | & & | \end{pmatrix}, \qquad \mathbf{\Lambda} \;=\; \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_d \end{pmatrix}.$$

Then $\mathbf{Q}$ is orthogonal and

$$\mathbf{A} \;=\; \mathbf{Q\Lambda Q}^\mathsf{T}, \qquad \mathbf{A} \;=\; \sum_{i=1}^{d} \lambda_i \mathbf{q}_i \mathbf{q}_i^\mathsf{T}.$$

Proof: The proof that an orthonormal eigenbasis exists is given in the optional appendix (Section 10). Here we verify the two displayed matrix identities. The eigenvector equations give

$$\mathbf{AQ} \;=\; \begin{pmatrix} \mathbf{Aq}_1 & \cdots & \mathbf{Aq}_d \end{pmatrix} \;=\; \begin{pmatrix} \lambda_1\mathbf{q}_1 & \cdots & \lambda_d\mathbf{q}_d \end{pmatrix} \;=\; \mathbf{Q\Lambda}.$$

Since $\mathbf{Q}$ is orthogonal, Lecture 5 gives $\mathbf{QQ}^\mathsf{T} = \mathbf{I}_d$. Multiplying on the right by $\mathbf{Q}^\mathsf{T}$ therefore gives $\mathbf{A} = \mathbf{Q\Lambda Q}^\mathsf{T}$. To obtain the second identity, expand this product using the columns of $\mathbf{Q}$ and the corresponding rows of $\mathbf{Q}^\mathsf{T}$:

$$\mathbf{Q\Lambda Q}^\mathsf{T} \;=\; \begin{pmatrix} \lambda_1\mathbf{q}_1 & \cdots & \lambda_d\mathbf{q}_d \end{pmatrix} \begin{pmatrix} \mathbf{q}_1^\mathsf{T} \\ \vdots \\ \mathbf{q}_d^\mathsf{T} \end{pmatrix} \;=\; \sum_{i=1}^{d} \lambda_i \mathbf{q}_i \mathbf{q}_i^\mathsf{T}.$$ □

The factorization describes three operations: $\mathbf{Q}^\mathsf{T}$ obtains orthonormal coordinates, $\mathbf{\Lambda}$ scales each coordinate and $\mathbf{Q}$ converts back to the original representation. If an eigenvalue repeats, its eigenspace can have different orthonormal bases. Gram-Schmidt can be used within that eigenspace without changing the eigenvalue of its vectors.

Example 8. The eigenvectors in Example 4 have lengths $\sqrt{14}$, $\sqrt{6}$ and $\sqrt{21}$, respectively. Normalizing them gives

$$\mathbf{q}_1 \;=\; \frac{1}{\sqrt{14}} \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}, \qquad \mathbf{q}_2 \;=\; \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \qquad \mathbf{q}_3 \;=\; \frac{1}{\sqrt{21}} \begin{pmatrix} -4 \\ 1 \\ 2 \end{pmatrix}.$$

They form an orthonormal basis, and

$$\mathbf{Q} \;=\; \begin{pmatrix} -\frac{1}{\sqrt{14}} & \frac{1}{\sqrt{6}} & -\frac{4}{\sqrt{21}} \\ \frac{2}{\sqrt{14}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{21}} \\ -\frac{3}{\sqrt{14}} & \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{21}} \end{pmatrix}, \qquad \mathbf{A} \;=\; \mathbf{Q} \begin{pmatrix} -28 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{Q}^{\mathsf{T}}.$$

For a vector $\mathbf{x} = \alpha_1 \mathbf{q}_1 + \alpha_2 \mathbf{q}_2 + \alpha_3 \mathbf{q}_3$, the transformation becomes $\mathbf{A}\mathbf{x} = -28\alpha_1 \mathbf{q}_1 + 18\alpha_2 \mathbf{q}_2$. In this basis, there is no mixing between the three coordinates.

Variation is measured using averages of squares, so it cannot be negative. The following matrix property will express this fact when we construct a matrix from the data.

Definition 9. For a symmetric matrix $\mathbf{A}$, the scalar $\mathbf{v}^{\mathsf{T}}\mathbf{A}\mathbf{v}$ is its quadratic form evaluated at $\mathbf{v}$. The matrix is positive semidefinite if $\mathbf{v}^{\mathsf{T}}\mathbf{A}\mathbf{v} \geq 0$ for every $\mathbf{v} \in \mathbb{R}^d$.

Proposition 10. A real symmetric matrix is positive semidefinite exactly when all its eigenvalues are nonnegative. More specifically, if $\mathbf{v} = \sum_{j=1}^{d} \alpha_j \mathbf{q}_j$ in an orthonormal eigenbasis, then

$$\mathbf{v}^{\mathsf{T}}\mathbf{A}\mathbf{v} \;=\; \sum_{j=1}^{d} \lambda_j \alpha_j^2.$$

Proof: By Theorem 7, $\mathbf{A}$ has an orthonormal eigenbasis $\mathbf{q}_1, \ldots, \mathbf{q}_d$ with real eigenvalues $\lambda_1, \ldots, \lambda_d$. Using $\mathbf{A}\mathbf{q}_j = \lambda_j \mathbf{q}_j$ and orthonormality,

$$\mathbf{v}^{\mathsf{T}}\mathbf{A}\mathbf{v} \;=\; \left\langle \sum_{i=1}^{d} \alpha_i \mathbf{q}_i, \sum_{j=1}^{d} \lambda_j \alpha_j \mathbf{q}_j \right\rangle \;=\; \sum_{i=1}^{d} \sum_{j=1}^{d} \alpha_i \lambda_j \alpha_j \left\langle \mathbf{q}_i, \mathbf{q}_j \right\rangle \;=\; \sum_{j=1}^{d} \lambda_j \alpha_j^2.$$

If every $\lambda_j \geq 0$, this is nonnegative for every $\mathbf{v}$. Conversely, if $\mathbf{A}$ is positive semidefinite, substituting $\mathbf{v} = \mathbf{q}_j$ gives $0 \leq \mathbf{q}_j^{\mathsf{T}}\mathbf{A}\mathbf{q}_j = \lambda_j$ for each j. □

Covariance captures how features vary together

For dimensionality reduction, we want to describe how datapoints differ from one another, rather than how far their common location lies from the origin. For instance, adding the same brightness offset to every image should not create a new direction of variation. We therefore subtract the mean datapoint before choosing directions. A covariance matrix then records how the centered features vary together.

Definition 11. Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ have rows $\mathbf{x}_1^{\mathsf{T}}, \ldots, \mathbf{x}_n^{\mathsf{T}}$, where $n \geq 2$. Its mean vector and centered datapoints are

$$\boldsymbol{\mu} \;=\; \frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i, \qquad \widetilde{\mathbf{x}}_i \;=\; \mathbf{x}_i - \boldsymbol{\mu}.$$

The centered data matrix is

$$\mathbf{X}_c \;=\; \mathbf{X} - \mathbf{1}\boldsymbol{\mu}^{\mathsf{T}} \in \mathbb{R}^{n \times d},$$

where $\mathbf{1} \in \mathbb{R}^n$ has every entry equal to 1. Its i-th row is $\widetilde{\mathbf{x}}_i^{\mathsf{T}}$.

The mean is the centroid from Lecture 1. By its definition, $\sum_{i=1}^n \widetilde{\mathbf{x}}_i = \sum_{i=1}^n \mathbf{x}_i - n\boldsymbol{\mu} = \mathbf{0}$. Thus each feature column of $\mathbf{X}_c$ has mean zero.

Definition 12. For real numbers $t_1, \ldots, t_n$ with mean $\bar{t} = \frac{1}{n}\sum_{i=1}^n t_i$, their empirical variance is

$$\mathrm{Var}(t_1, \ldots, t_n) = \frac{1}{n}\sum_{i=1}^n (t_i - \bar{t})^2.$$

The covariance matrix of a dataset is

$$\mathbf{C} = \frac{1}{n}\mathbf{X}_c^\mathsf{T}\mathbf{X}_c \in \mathbb{R}^{d\times d}.$$

Its (j, ℓ)-th entry is

$$C_{j\ell} = \frac{1}{n}\sum_{i=1}^n (x_{ij} - \mu_j)(x_{i\ell} - \mu_\ell).$$

The diagonal entry C_{jj} is the variance of feature j. An off-diagonal entry averages the products of the deviations of two features from their respective means. Some texts use $1/(n-1)$ instead of $1/n$ for covariance. We use $1/n$ throughout: it makes variance exactly the average squared deviation. Switching between these two conventions scales all eigenvalues by the same positive factor and does not change the principal-component directions.

Proposition 13. Every covariance matrix is symmetric and positive semidefinite.

Proof: The transpose rule from Lecture 2 gives

$$\mathbf{C}^\mathsf{T} = \frac{1}{n}(\mathbf{X}_c^\mathsf{T}\mathbf{X}_c)^\mathsf{T} = \frac{1}{n}\mathbf{X}_c^\mathsf{T}\mathbf{X}_c = \mathbf{C}.$$

Hence $\mathbf{C}$ is symmetric. For every $\mathbf{v} \in \mathbb{R}^d$,

$$\mathbf{v}^\mathsf{T}\mathbf{C}\mathbf{v} = \frac{1}{n}\mathbf{v}^\mathsf{T}\mathbf{X}_c^\mathsf{T}\mathbf{X}_c\mathbf{v} = \frac{1}{n}\|\mathbf{X}_c\mathbf{v}\|_2^2 \geq 0.$$

Hence $\mathbf{C}$ is positive semidefinite. □

Example 14. Consider four datapoints in $\mathbb{R}^2$, stored as rows:

$$\mathbf{X} = \begin{pmatrix} 16 & 24 \\ 14 & 26 \\ 4 & 16 \\ 6 & 14 \end{pmatrix}.$$

Their mean and centered data matrix are

$$\boldsymbol{\mu} = \begin{pmatrix} 10 \\ 20 \end{pmatrix}, \qquad \mathbf{X}_c = \begin{pmatrix} 6 & 4 \\ 4 & 6 \\ -6 & -4 \\ -4 & -6 \end{pmatrix}.$$

Therefore

$$\mathbf{C} \;=\; \frac{1}{4}\begin{pmatrix} 6 & 4 & -6 & -4 \\ 4 & 6 & -4 & -6 \end{pmatrix}\begin{pmatrix} 6 & 4 \\ 4 & 6 \\ -6 & -4 \\ -4 & -6 \end{pmatrix} \;=\; \frac{1}{4}\begin{pmatrix} 104 & 96 \\ 96 & 104 \end{pmatrix} \;=\; \begin{pmatrix} 26 & 24 \\ 24 & 26 \end{pmatrix}.$$

Each original feature has variance 26. The covariance 24 reflects that the two features tend to be above their means together and below their means together. The vectors $(1,1)^\mathsf{T}$ and $(1,-1)^\mathsf{T}$ are eigenvectors of $\mathbf{C}$ with eigenvalues 50 and 2, respectively.

Now consider a second dataset with the same mean $\boldsymbol{\mu} = (10,20)^\mathsf{T}$:

$$\mathbf{X}' \;=\; \begin{pmatrix} 16 & 14 \\ 14 & 16 \\ 4 & 26 \\ 6 & 24 \end{pmatrix}, \qquad \mathbf{X}'_c \;=\; \begin{pmatrix} 6 & -6 \\ 4 & -4 \\ -6 & 6 \\ -4 & 4 \end{pmatrix}.$$

Its covariance matrix is $\mathbf{C}' = \frac{1}{4}(\mathbf{X}'_c)^\mathsf{T}\mathbf{X}'_c = \begin{pmatrix} 26 & -26 \\ -26 & 26 \end{pmatrix}$. Both features still have variance 26, but their covariance is now -26. The negative off-diagonal entries reflect that when the first feature is above its mean, the second is below its mean, and vice versa. Here the relationship is exact: the two features sum to 30 for every datapoint, so their centered values are negatives of one another.

The first principal component retains the most variance

Suppose we can store only one number for each centered datapoint. Choosing a unit vector $\mathbf{q}$ lets us store the coordinate $\langle \mathbf{q}, \widetilde{\mathbf{x}}_i \rangle$ along that direction. If all these coordinates are nearly equal, the representation hides most differences between datapoints. PCA instead chooses a direction along which the coordinates have maximum variance.

Proposition 15. For a unit vector $\mathbf{q} \in \mathbb{R}^d$, let $t_i = \langle \mathbf{q}, \widetilde{\mathbf{x}}_i \rangle$. These coordinates have mean zero, and their variance is

$$\mathrm{Var}(t_1, \ldots, t_n) \;=\; \frac{1}{n}\sum_{i=1}^{n} \langle \mathbf{q}, \widetilde{\mathbf{x}}_i \rangle^2 \;=\; \mathbf{q}^\mathsf{T}\mathbf{C}\mathbf{q}.$$

Proof: Since the centered datapoints sum to zero,

$$\frac{1}{n}\sum_{i=1}^{n} t_i \;=\; \left\langle \mathbf{q}, \frac{1}{n}\sum_{i=1}^{n} \widetilde{\mathbf{x}}_i \right\rangle \;=\; 0.$$

The vector of all coordinates is $\mathbf{t} = \mathbf{X}_c\mathbf{q}$, because row i of $\mathbf{X}_c$ is $\widetilde{\mathbf{x}}_i^\mathsf{T}$. Consequently,

$$\frac{1}{n}\sum_{i=1}^{n} t_i^2 \;=\; \frac{1}{n}\|\mathbf{X}_c\mathbf{q}\|_2^2 \;=\; \mathbf{q}^\mathsf{T}\left(\frac{1}{n}\mathbf{X}_c^\mathsf{T}\mathbf{X}_c\right)\mathbf{q} \;=\; \mathbf{q}^\mathsf{T}\mathbf{C}\mathbf{q}. \qquad \square$$

The unit-length condition prevents an artificial increase in variance: replacing $\mathbf{q}$ by $a\mathbf{q}$ multiplies all scores by a and their variance by a^2. We want to compare directions at the same scale.

Definition 16. A first principal-component direction is a unit vector $\mathbf{q}$ maximizing $\mathbf{q}^{\mathsf{T}}\mathbf{C}\mathbf{q}$. The numbers $\langle \mathbf{q}, \widetilde{\mathbf{x}}_i \rangle$ are the corresponding principal-component scores of the datapoints.

Theorem 17. Let $\mathbf{C}$ have an orthonormal eigenbasis $\mathbf{q}_1, \ldots, \mathbf{q}_d$, ordered so that their eigenvalues satisfy $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$. Then

$$\max_{\|\mathbf{q}\|_2 = 1} \mathbf{q}^{\mathsf{T}}\mathbf{C}\mathbf{q} \;=\; \lambda_1.$$

The unit vectors in the eigenspace of λ_1 are exactly the maximizers. In particular, $\mathbf{q}_1$ is a first principal-component direction.

Proof: Write a unit vector as $\mathbf{q} = \sum_{j=1}^{d} \alpha_j \mathbf{q}_j$. Orthonormality gives $\sum_{j=1}^{d} \alpha_j^2 = 1$. By Proposition 10,

$$\mathbf{q}^{\mathsf{T}}\mathbf{C}\mathbf{q} \;=\; \sum_{j=1}^{d} \lambda_j \alpha_j^2 \;\leq\; \lambda_1 \sum_{j=1}^{d} \alpha_j^2 \;=\; \lambda_1.$$

Equality holds exactly when $\alpha_j = 0$ for every j with $\lambda_j < \lambda_1$. These are precisely the unit vectors in the eigenspace of λ_1. Choosing $\mathbf{q} = \mathbf{q}_1$ attains the bound. □

Example 18. For this and the following PCA examples, consider six datapoints in $\mathbb{R}^4$. Their data matrix, mean and centered data matrix are

$$\mathbf{X} \;=\; \begin{pmatrix} 38 & 43 & 41 & 50 \\ 20 & 31 & 53 & 68 \\ 26 & 40 & 50 & 56 \\ 19 & 30 & 22 & 37 \\ 7 & 12 & 40 & 49 \\ 10 & 24 & 34 & 40 \end{pmatrix}, \qquad \boldsymbol{\mu} \;=\; \begin{pmatrix} 20 \\ 30 \\ 40 \\ 50 \end{pmatrix}, \qquad \mathbf{X}_c \;=\; \begin{pmatrix} 18 & 13 & 1 & 0 \\ 0 & 1 & 13 & 18 \\ 6 & 10 & 10 & 6 \\ -1 & 0 & -18 & -13 \\ -13 & -18 & 0 & -1 \\ -10 & -6 & -6 & -10 \end{pmatrix}.$$

The covariance matrix is

$$\mathbf{C} \;=\; \frac{1}{6}\mathbf{X}_c^{\mathsf{T}}\mathbf{X}_c \;=\; \begin{pmatrix} 105 & 98 & 26 & 27 \\ 98 & 105 & 27 & 26 \\ 26 & 27 & 105 & 98 \\ 27 & 26 & 98 & 105 \end{pmatrix}.$$

Its eigenvalues, in decreasing order, are 256, 150, 8 and 6. An orthonormal eigenbasis is given in Example 21. The first principal-component direction and its six scores are

$$\mathbf{q}_1 \;=\; \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \qquad \mathbf{t} \;=\; \mathbf{X}_c\mathbf{q}_1 \;=\; \begin{pmatrix} 16 \\ 16 \\ 16 \\ -16 \\ -16 \\ -16 \end{pmatrix}.$$

The scores have mean zero and variance

$$\frac{1}{6}\sum_{i=1}^{6} t_i^2 \;=\; \frac{1}{6} \times (6 \times 16^2) \;=\; 256.$$

In contrast, each original centered feature has variance 105. Combining features in the first principal-component direction therefore retains more variation than any one original feature alone.

Changing $\mathbf{q}_1$ to $-\mathbf{q}_1$ reverses every score but leaves the selected line and all reconstruction distances unchanged. If the largest eigenvalue repeats, there can also be several different maximizing directions, not just this sign ambiguity.

Several principal components retain a subspace

To keep two or more coordinates, simply repeating the first direction would store the same information repeatedly. Instead, we require mutually orthogonal directions. The second direction captures as much variation as possible subject to being orthogonal to the first, and we continue in the same way. The ordered eigenvectors of the covariance matrix provide these directions.

Definition 19. Let $\mathbf{q}_1, \dots, \mathbf{q}_d$ be an orthonormal eigenbasis of $\mathbf{C}$, ordered by $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$. For $1 \leq k \leq d$, define

$$\mathbf{Q}_k \;=\; \begin{pmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_k \\ | & & | \end{pmatrix} \in \mathbb{R}^{d \times k}.$$

Principal component analysis retains the k-dimensional representation

$$\mathbf{z}_i \;=\; \mathbf{Q}_k^{\mathsf{T}}(\mathbf{x}_i - \boldsymbol{\mu}) \in \mathbb{R}^k.$$

The corresponding batch of representations is

$$\mathbf{Z} \;=\; \mathbf{X}_c \mathbf{Q}_k \in \mathbb{R}^{n \times k},$$

whose i-th row is $\mathbf{z}_i^{\mathsf{T}}$. The retained subspace is $\mathcal{S}_k = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$.

For a general centered datapoint, $\mathbf{z}_i$ gives the coordinates of its projection onto $\mathcal{S}_k$, not an exact representation of the whole datapoint. This distinction is the same one made for orthogonal projection.

Proposition 20. For each $j = 1, \dots, d$, the vector $\mathbf{q}_j$ maximizes variance among unit vectors orthogonal to $\mathbf{q}_1, \dots, \mathbf{q}_{j-1}$. The maximum is λ_j. Moreover, the retained scores have mean zero and covariance

$$\frac{1}{n}\mathbf{Z}^{\mathsf{T}}\mathbf{Z} \;=\; \mathbf{Q}_k^{\mathsf{T}}\mathbf{C}\mathbf{Q}_k \;=\; \operatorname{diag}(\lambda_1, \dots, \lambda_k).$$

Proof: Any unit vector orthogonal to the first $j-1$ eigenvectors has an expansion $\mathbf{q} = \sum_{\ell=j}^{d} \alpha_\ell \mathbf{q}_\ell$ with $\sum_{\ell=j}^{d} \alpha_\ell^2 = 1$. Therefore

$$\mathbf{q}^{\mathsf{T}}\mathbf{C}\mathbf{q} \;=\; \sum_{\ell=j}^{d} \lambda_\ell \alpha_\ell^2 \;\leq\; \lambda_j,$$

with equality for $\mathbf{q} = \mathbf{q}_j$.

The scores have mean zero because $\frac{1}{n}\sum_i \mathbf{z}_i = \mathbf{Q}_k^{\mathsf{T}} \frac{1}{n}\sum_i \widetilde{\mathbf{x}}_i = \mathbf{0}$. The batch formula gives

$$\frac{1}{n}\mathbf{Z}^{\mathsf{T}}\mathbf{Z} \;=\; \mathbf{Q}_k^{\mathsf{T}} \left(\frac{1}{n}\mathbf{X}_c^{\mathsf{T}}\mathbf{X}_c\right) \mathbf{Q}_k \;=\; \mathbf{Q}_k^{\mathsf{T}}\mathbf{C}\mathbf{Q}_k.$$

Its (j, ℓ)-th entry is

$$\mathbf{q}_j^{\mathsf{T}}\mathbf{C}\mathbf{q}_\ell \;=\; \lambda_\ell \langle \mathbf{q}_j, \mathbf{q}_\ell \rangle \;=\; \begin{cases} \lambda_j, & j = \ell, \\ 0, & j \neq \ell. \end{cases}$$

This is the stated diagonal matrix. □

Thus each retained coordinate contributes variance λ_j, and the sum of their variances is $\lambda_1 + \cdots + \lambda_k$. Distinct score coordinates have zero empirical covariance; we say that they are uncorrelated.

Example 21. For the six datapoints in Example 18, an ordered orthonormal eigenbasis of $\mathbf{C}$ is

eigenvector	eigenvalue
$\mathbf{q}_1 = (1,1,1,1)^\mathsf{T}/2$	256
$\mathbf{q}_2 = (1,1,-1,-1)^\mathsf{T}/2$	150
$\mathbf{q}_3 = (1,-1,-1,1)^\mathsf{T}/2$	8
$\mathbf{q}_4 = (1,-1,1,-1)^\mathsf{T}/2$	6

Multiplying $\mathbf{C}$ by each displayed vector verifies its eigenvalue equation. Keeping the first two directions gives

$$\mathbf{Q}_2 \;=\; \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \\ 1 & -1 \end{pmatrix}, \qquad \mathbf{Z} \;=\; \mathbf{X}_c\mathbf{Q}_2 \;=\; \begin{pmatrix} 16 & 15 \\ 16 & -15 \\ 16 & 0 \\ -16 & 15 \\ -16 & -15 \\ -16 & 0 \end{pmatrix}.$$

For example, the first datapoint is represented by $\mathbf{z}_1 = (16, 15)^\mathsf{T}$. The two retained score columns have variances 256 and 150. The third and fourth principal components, whose variances are only 8 and 6, are omitted.

PCA minimizes reconstruction error

Keeping variance is useful only if it gives a meaningful approximation of the original data. Lecture 5 tells us how to reconstruct a datapoint from its retained orthonormal coordinates: multiply by the basis matrix. We must also restore the mean that we subtracted. We now show that, among all subspaces of the same dimension, the principal-component subspace minimizes the average squared reconstruction error.

Definition 22. For a matrix $\mathbf{U} \in \mathbb{R}^{d\times k}$ with orthonormal columns, the reconstruction of $\mathbf{x}_i$ using the corresponding subspace is

$$\widehat{\mathbf{x}}_i \;=\; \boldsymbol{\mu} + \mathbf{U}\mathbf{U}^\mathsf{T}(\mathbf{x}_i - \boldsymbol{\mu}).$$

Its average squared reconstruction error on the training dataset is

$$E(\mathbf{U}) \;=\; \frac{1}{n}\sum_{i=1}^{n} \|\mathbf{x}_i - \widehat{\mathbf{x}}_i\|_2^2.$$

Theorem 23. Let $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$ be the eigenvalues of the covariance matrix $\mathbf{C}$. For every $\mathbf{U} = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_k \end{pmatrix}$ with orthonormal columns,

$$E(\mathbf{U}) \;=\; \sum_{\ell=1}^{d} \lambda_\ell - \sum_{j=1}^{k} \mathbf{u}_j^\mathsf{T}\mathbf{C}\mathbf{u}_j \;\geq\; \sum_{\ell=k+1}^{d} \lambda_\ell.$$

Equality is attained by $\mathbf{U} = \mathbf{Q}_k$. Thus PCA both maximizes the sum of retained variances and minimizes the average squared reconstruction error among k-dimensional subspaces of the data space.

Proof: For each centered datapoint, its projection $\mathbf{U}\mathbf{U}^\mathsf{T}\widetilde{\mathbf{x}}_i$ is orthogonal to the residual. The projection theorem from Lecture 5 therefore gives

$$\|\widetilde{\mathbf{x}}_i\|_2^2 \;=\; \|\mathbf{U}\mathbf{U}^\mathsf{T}\widetilde{\mathbf{x}}_i\|_2^2 + \|\widetilde{\mathbf{x}}_i - \mathbf{U}\mathbf{U}^\mathsf{T}\widetilde{\mathbf{x}}_i\|_2^2.$$

By orthonormality, the first term on the right is $\sum_{j=1}^k \langle \mathbf{u}_j, \widetilde{\mathbf{x}}_i\rangle^2$. Also, expanding each centered datapoint in the full orthonormal eigenbasis gives

$$\frac{1}{n}\sum_{i=1}^n \|\widetilde{\mathbf{x}}_i\|_2^2 \;=\; \sum_{\ell=1}^d \left(\frac{1}{n}\sum_{i=1}^n \langle \mathbf{q}_\ell, \widetilde{\mathbf{x}}_i\rangle^2\right) \;=\; \sum_{\ell=1}^d \lambda_\ell.$$

Averaging the projection identity and using Proposition 15 gives

$$E(\mathbf{U}) \;=\; \sum_{\ell=1}^d \lambda_\ell - \sum_{j=1}^k \mathbf{u}_j^\mathsf{T}\mathbf{C}\mathbf{u}_j.$$

It remains to bound the retained variance for an arbitrary orthonormal choice of $\mathbf{U}$. Define

$$a_\ell \;=\; \sum_{j=1}^k \langle \mathbf{q}_\ell, \mathbf{u}_j\rangle^2.$$

This is the squared norm of the projection of the unit vector $\mathbf{q}_\ell$ onto $\mathrm{Col}(\mathbf{U})$, so $0 \le a_\ell \le 1$. Moreover,

$$\sum_{\ell=1}^d a_\ell \;=\; \sum_{j=1}^k\sum_{\ell=1}^d \langle \mathbf{q}_\ell, \mathbf{u}_j\rangle^2 \;=\; \sum_{j=1}^k \|\mathbf{u}_j\|_2^2 \;=\; k.$$

In particular, $\sum_{\ell=k+1}^d a_\ell = \sum_{\ell=1}^k (1-a_\ell)$. Expanding the quadratic forms in the eigenbasis now gives

$$\sum_{j=1}^k \mathbf{u}_j^\mathsf{T}\mathbf{C}\mathbf{u}_j \;=\; \sum_{\ell=1}^d \lambda_\ell a_\ell \;\le\; \sum_{\ell=1}^k \lambda_\ell a_\ell + \lambda_k \sum_{\ell=k+1}^d a_\ell \;=\; \sum_{\ell=1}^k \big(\lambda_\ell a_\ell + \lambda_k(1-a_\ell)\big) \;\le\; \sum_{\ell=1}^k \lambda_\ell.$$

The first inequality uses $\lambda_\ell \le \lambda_k$ for $\ell > k$. The last uses $\lambda_k \le \lambda_\ell$ and $1 - a_\ell \ge 0$ for $\ell \le k$. Choosing $\mathbf{U} = \mathbf{Q}_k$ retains exactly $\lambda_1 + \cdots + \lambda_k$, so it attains the error bound. □

Corollary 24. For any fixed subspace, choosing the dataset mean as the reconstruction offset minimizes the average squared error among all offsets. Consequently, PCA also gives a best k-dimensional affine subspace (a translated linear subspace) for the original, uncentered data.

Proof: Fix an orthonormal basis matrix $\mathbf{U}$ and write $\mathbf{P} = \mathbf{U}\mathbf{U}^\mathsf{T}$. Using an offset $\mathbf{a}$ gives reconstruction $\mathbf{a} + \mathbf{P}(\mathbf{x}_i - \mathbf{a})$ and residual

$$(\mathbf{I}_d - \mathbf{P})(\mathbf{x}_i - \mathbf{a}) \;=\; (\mathbf{I}_d - \mathbf{P})\widetilde{\mathbf{x}}_i + (\mathbf{I}_d - \mathbf{P})(\boldsymbol{\mu} - \mathbf{a}).$$

When we square and average, the cross term vanishes because $\sum_i \widetilde{\mathbf{x}}_i = \mathbf{0}$. Thus the average error is

$$E(\mathbf{U}) + \|(\mathbf{I}_d - \mathbf{P})(\boldsymbol{\mu} - \mathbf{a})\|_2^2 \;\ge\; E(\mathbf{U}).$$

Taking $\mathbf{a} = \boldsymbol{\mu}$ attains equality. We can therefore choose the mean offset first and then apply Theorem 23 to choose the subspace. □

Example 25. Use the six datapoints in Example 18 and retain the two coordinates from Example 21. For the first datapoint,

$$\widehat{\mathbf{x}}_1 \;=\; \boldsymbol{\mu} + \mathbf{Q}_2\mathbf{z}_1 \;=\; \boldsymbol{\mu} + 16\mathbf{q}_1 + 15\mathbf{q}_2 \;=\; \frac{1}{2}\begin{pmatrix} 71 \\ 91 \\ 81 \\ 101 \end{pmatrix}.$$

The full reconstruction and residual matrices are

$$\widehat{\mathbf{X}} \;=\; \frac{1}{2}\begin{pmatrix} 71 & 91 & 81 & 101 \\ 41 & 61 & 111 & 131 \\ 56 & 76 & 96 & 116 \\ 39 & 59 & 49 & 69 \\ 9 & 29 & 79 & 99 \\ 24 & 44 & 64 & 84 \end{pmatrix}, \qquad \mathbf{X} - \widehat{\mathbf{X}} \;=\; \frac{1}{2}\begin{pmatrix} 5 & -5 & 1 & -1 \\ -1 & 1 & -5 & 5 \\ -4 & 4 & 4 & -4 \\ -1 & 1 & -5 & 5 \\ 5 & -5 & 1 & -1 \\ -4 & 4 & 4 & -4 \end{pmatrix}.$$

The six squared reconstruction errors are 13, 13, 16, 13, 13 and 16. Their average is

$$E(\mathbf{Q}_2) \;=\; \frac{1}{6}(13 + 13 + 16 + 13 + 13 + 16) \;=\; 14 \;=\; \lambda_3 + \lambda_4.$$

Thus the average error equals the sum of the discarded eigenvalues, although an individual datapoint's error need not equal this average. Retaining only the first principal component would instead give average error $150 + 8 + 6 = 164$.

Corollary 26. If $\mathbf{X}_c$ has rank r, its covariance matrix has exactly r positive eigenvalues. Moreover,

$$\operatorname{rank}(\mathbf{C}) \;=\; \operatorname{rank}(\mathbf{X}_c) \;=\; r \;\leq\; \min(d, n-1).$$

Retaining all these directions reconstructs every training datapoint exactly.

Proof: The matrices $\mathbf{C}$ and $\mathbf{X}_c$ have the same null space. Indeed, $\mathbf{X}_c\mathbf{v} = \mathbf{0}$ implies $\mathbf{C}\mathbf{v} = \frac{1}{n}\mathbf{X}_c^{\mathsf{T}}\mathbf{X}_c\mathbf{v} = \mathbf{0}$. Conversely, $\mathbf{C}\mathbf{v} = \mathbf{0}$ implies

$$0 \;=\; \mathbf{v}^{\mathsf{T}}\mathbf{C}\mathbf{v} \;=\; \frac{1}{n}\|\mathbf{X}_c\mathbf{v}\|_2^2,$$

so $\mathbf{X}_c\mathbf{v} = \mathbf{0}$. Both matrices have d columns, so rank-nullity gives $\operatorname{rank}(\mathbf{C}) = \operatorname{rank}(\mathbf{X}_c) = r$. Since the centered rows sum to zero, every column of $\mathbf{X}_c$ lies in the subspace $\{\mathbf{y} \in \mathbb{R}^n : \sum_{i=1}^n y_i = 0\}$. This subspace has dimension $n-1$, because its first $n-1$ coordinates can be chosen freely and determine the last coordinate. Thus $r \leq n-1$. Also, $r \leq d$ because $\mathbf{X}_c$ has d columns, so $r \leq \min(d, n-1)$.

In an eigenbasis, the null space of $\mathbf{C}$ is spanned precisely by the eigenvectors with eigenvalue zero. Hence exactly r eigenvalues are positive. Retaining their directions leaves a sum of discarded eigenvalues equal to zero, so Theorem 23 gives zero average error. Each squared error is nonnegative, so each must be zero. If $r = 0$, all centered datapoints are zero and the mean alone already reconstructs the dataset. □

Thus PCA includes the exact low-dimensional representation discussed in Lecture 4, but also handles approximate structure by discarding directions with small positive eigenvalues. When eigenvalues tie at the cutoff, there can be several equally good retained subspaces.

The bound in Corollary 26 also connects the number of samples to the dimension. When $d \geq n$, the centered training datapoints necessarily lie in a subspace of dimension at most $n-1$, even if the observations consist entirely of noise. Exact reconstruction using $n-1$ components is therefore not, by itself, evidence of meaningful low-dimensional structure. The useful question is whether a substantially smaller number of components retains relevant structure and gives a suitable representation of new datapoints.

Using PCA on a dataset

The formulas now give a complete data-processing procedure. We learn a mean and a set of directions from the training data, then use that same mean and those same directions to represent new datapoints. The eigenvalues quantify the training variation retained by each possible choice of representation size.

Definition 27. If $\sum_{j=1}^{d} \lambda_j > 0$, the explained variance ratio of a k-component PCA representation is

$$\rho_k = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{d} \lambda_j}.$$

The numerator is the retained variance. The denominator is the total centered variance $\frac{1}{n}\sum_i \|\mathbf{x}_i - \boldsymbol{\mu}\|_2^2$.

One way to choose k is to select the smallest value for which ρ_k reaches a chosen threshold.

Example 28. For the six datapoints in Example 18, keeping the first two principal components retains variance $256+150=406$ out of a total $256+150+8+6=420$. The explained variance ratio is

$$\rho_2 = \frac{256+150}{256+150+8+6} = \frac{406}{420} \approx 0.9667.$$

The two-dimensional representation therefore retains about 96.67% of the training variance. It meets a 95% variance threshold, but not a 99% threshold. The discarded fraction is $1/30$, corresponding to the average squared reconstruction error 14 from Example 25.

Algorithm 1 PCA with k retained components

Require: Training data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and $1 \leq k \leq d$

1: Compute the training mean $\boldsymbol{\mu} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i$
2: Center the data: $\mathbf{X}_c = \mathbf{X} - \mathbf{1}\boldsymbol{\mu}^{\mathsf{T}}$
3: Optionally scale the centered features of $\mathbf{X}_c$ according to the application; retain the scaling parameters for future datapoints
4: Form $\mathbf{C} = \frac{1}{n}\mathbf{X}_c^{\mathsf{T}}\mathbf{X}_c$
5: Obtain orthonormal eigenvectors of $\mathbf{C}$, ordered by decreasing eigenvalue
6: Form $\mathbf{Q}_k$ from the first k eigenvectors
7: Compute the training scores $\mathbf{Z} = \mathbf{X}_c\mathbf{Q}_k$
8: Retain $\boldsymbol{\mu}$, $\mathbf{Q}_k$ and any feature-scaling parameters to transform future datapoints

In practice, numerical software computes the covariance matrix's eigenvalues and an orthonormal eigenbasis; we do not need to find them by hand.

Without optional feature scaling, encoding and reconstruction of a new datapoint $\mathbf{x}$ use

$$\mathbf{z} \;=\; \mathbf{Q}_k^{\mathsf{T}}(\mathbf{x}-\boldsymbol{\mu}), \qquad \widehat{\mathbf{x}} \;=\; \boldsymbol{\mu}+\mathbf{Q}_k\mathbf{z}.$$

For the training batch, the reconstruction is

$$\widehat{\mathbf{X}} \;=\; \mathbf{1}\boldsymbol{\mu}^{\mathsf{T}}+\mathbf{Z}\mathbf{Q}_k^{\mathsf{T}} \in \mathbb{R}^{n\times d}.$$

When optional feature scaling is used, apply the fitted scaling to centered datapoints before encoding and undo it before adding the mean during reconstruction. The variance and reconstruction-error optimality results then apply in the scaled feature space.

The mean and directions must be fitted using the training data, rather than recomputed using the test data. Otherwise, the fitted representation would use information that was meant to be held out for evaluation.

PCA has several practical uses. A smaller representation can reduce storage and the cost of subsequent models. Removing low-variance directions can also reduce noise when the noise is concentrated in those directions. Neither interpretation means that high variance is always useful for prediction: PCA does not use class labels, and a low-variance direction may be essential for distinguishing two classes.

Feature units also matter. Changing a measurement from metres to millimetres multiplies its variance by 10^6, potentially changing the leading directions. Features should therefore be scaled according to the application before fitting PCA; the same scaling must then be applied to new datapoints.

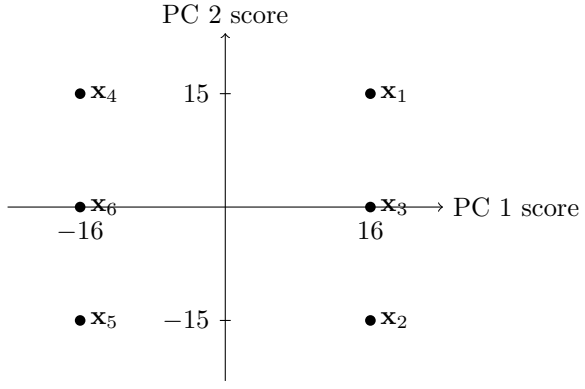
Visualizing high-dimensional data

A datapoint in $\mathbb{R}^d$ cannot be plotted directly when $d > 3$, but its first two or three PCA scores can. For a two-dimensional plot, we place each datapoint at

$$\mathbf{z}_i \;=\; \begin{pmatrix} \langle \mathbf{q}_1, \mathbf{x}_i-\boldsymbol{\mu}\rangle \\ \langle \mathbf{q}_2, \mathbf{x}_i-\boldsymbol{\mu}\rangle \end{pmatrix}.$$

A three-dimensional plot also includes the score along $\mathbf{q}_3$. The plot axes represent linear combinations of the original features, rather than simply two or three selected features. Such plots can help us look for groups, unusual observations or broad patterns in the data.

Example 29. The dataset in Example 21 has $n = 6$ datapoints in $\mathbb{R}^4$. Its first two PCA scores give the following two-dimensional visualization. Each label identifies the original datapoint.



The plot shows two groups separated along the first principal component, with variation within each group along the second. By Example 28, these two directions retain about 96.67% of the training variance. The variation along $\mathbf{q}_3$ and $\mathbf{q}_4$ is not visible.

A PCA plot is only a partial view of the data. The explained variance ratio ρ_2 or ρ_3 tells us how much training variance the plotted directions retain; two or three components need not capture most of it. Datapoints that overlap in the plot may differ substantially in discarded directions. Consequently, the plot need not preserve all distances or reveal every distinction between classes.

This lecture obtained PCA from the eigenvectors of $\mathbf{X}_c^\mathsf{T}\mathbf{X}_c$. In the next lecture, *singular value decomposition* (SVD) will connect these directions directly to the rectangular data matrix $\mathbf{X}_c$, and extend the same idea to approximating general matrices by low-rank matrices.

Optional appendix: Proof of the spectral theorem

This appendix supplies the existence argument for Theorem 7. We first find one eigenvector by maximizing a quadratic form, then use induction to find the remaining orthonormal eigenvectors. The proof uses the standard *extreme-value theorem* from analysis.

We argue by induction on d. For $d = 1$, the vector (1) is an orthonormal basis and an eigenvector of $\mathbf{A} = (a_{11})$ with real eigenvalue a_{11}. Let $d \geq 2$ and assume the statement holds in dimension $d - 1$.

Step 1: Find one eigenvector. The unit sphere in $\mathbb{R}^d$ is nonempty, closed and bounded, hence compact by the Heine–Borel theorem. The function $\mathbf{x} \mapsto \mathbf{x}^\mathsf{T}\mathbf{A}\mathbf{x}$ is continuous because it is a polynomial in the entries of $\mathbf{x}$. By the extreme-value theorem, it attains a maximum at some unit vector $\mathbf{q}_1$. Set

$$\lambda_1 \;=\; \mathbf{q}_1^\mathsf{T}\mathbf{A}\mathbf{q}_1 \in \mathbb{R}.$$

We show that $\mathbf{A}\mathbf{q}_1 = \lambda_1\mathbf{q}_1$ without assuming an eigenbasis.

Let $\mathbf{u} \perp \mathbf{q}_1$. For every $t \in \mathbb{R}$, the vector

$$\mathbf{q}(t) \;=\; \frac{\mathbf{q}_1 + t\mathbf{u}}{\sqrt{1 + t^2\|\mathbf{u}\|_2^2}}$$

has unit length: orthogonality gives $\|\mathbf{q}_1 + t\mathbf{u}\|_2^2 = 1 + t^2\|\mathbf{u}\|_2^2$. Maximality therefore gives $\mathbf{q}(t)^\mathsf{T}\mathbf{A}\mathbf{q}(t) \leq \lambda_1$. Expanding the numerator and using $\mathbf{A}^\mathsf{T} = \mathbf{A}$ yields

$$\frac{\lambda_1 + 2t\,\langle \mathbf{u}, \mathbf{A}\mathbf{q}_1\rangle + t^2\mathbf{u}^\mathsf{T}\mathbf{A}\mathbf{u}}{1 + t^2\|\mathbf{u}\|_2^2} \;\leq\; \lambda_1.$$

Multiplying by the positive denominator and cancelling λ_1 gives

$$2t\,\langle \mathbf{u}, \mathbf{A}\mathbf{q}_1\rangle + t^2\big(\mathbf{u}^\mathsf{T}\mathbf{A}\mathbf{u} - \lambda_1\|\mathbf{u}\|_2^2\big) \;\leq\; 0.$$

For $t > 0$, divide by t and let t tend to zero to obtain $\langle \mathbf{u}, \mathbf{A}\mathbf{q}_1\rangle \leq 0$. For $t < 0$, division reverses the inequality, and the same limit gives $\langle \mathbf{u}, \mathbf{A}\mathbf{q}_1\rangle \geq 0$. Thus $\langle \mathbf{u}, \mathbf{A}\mathbf{q}_1\rangle = 0$ for every $\mathbf{u} \perp \mathbf{q}_1$.

To conclude, put $\mathbf{r} = \mathbf{A}\mathbf{q}_1 - \lambda_1\mathbf{q}_1$. The definition of λ_1 gives

$$\langle \mathbf{q}_1, \mathbf{r}\rangle \;=\; \mathbf{q}_1^\mathsf{T}\mathbf{A}\mathbf{q}_1 - \lambda_1\|\mathbf{q}_1\|_2^2 \;=\; 0.$$

We may therefore take $\mathbf{u} = \mathbf{r}$ in the preceding conclusion. Consequently,

$$\|\mathbf{r}\|_2^2 \;=\; \langle \mathbf{r}, \mathbf{A}\mathbf{q}_1 - \lambda_1\mathbf{q}_1\rangle \;=\; 0,$$

so $\mathbf{r} = \mathbf{0}$ and $\mathbf{q}_1$ is an eigenvector.

Step 2: Restrict to the perpendicular subspace. Extend $\mathbf{q}_1$ to a basis of $\mathbb{R}^d$ and apply Gram-Schmidt. Since $\mathbf{q}_1$ already has unit length, it remains the first vector of the resulting orthonormal basis. Stack the other basis vectors as the columns of $\mathbf{U} \in \mathbb{R}^{d\times(d-1)}$. Then

$$\mathbf{U}^\mathsf{T}\mathbf{U} \;=\; \mathbf{I}_{d-1}, \qquad \mathbf{q}_1^\mathsf{T}\mathbf{U} \;=\; \mathbf{0}^\mathsf{T}.$$

The columns of $\mathbf{U}$ span exactly the vectors perpendicular to $\mathbf{q}_1$. For any such vector $\mathbf{x}$, symmetry and the eigenvector equation give

$$\langle \mathbf{q}_1, \mathbf{A}\mathbf{x} \rangle \;=\; \langle \mathbf{A}\mathbf{q}_1, \mathbf{x} \rangle \;=\; \lambda_1 \langle \mathbf{q}_1, \mathbf{x} \rangle \;=\; 0.$$

Thus $\mathbf{A}\mathbf{x}$ is also perpendicular to $\mathbf{q}_1$. In particular, each column of $\mathbf{AU}$ lies in $\mathrm{Col}(\mathbf{U})$. The orthonormal-coordinate formula from Lecture 5 therefore gives

$$\mathbf{AU} \;=\; \mathbf{UB}, \qquad \mathbf{B} \;=\; \mathbf{U}^{\mathsf{T}}\mathbf{AU} \in \mathbb{R}^{(d-1)\times(d-1)}.$$

This smaller matrix is real symmetric, since

$$\mathbf{B}^{\mathsf{T}} \;=\; \mathbf{U}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{U} \;=\; \mathbf{U}^{\mathsf{T}}\mathbf{AU} \;=\; \mathbf{B}.$$

Step 3: Apply induction. By the induction hypothesis, $\mathbf{B}$ has an orthonormal eigenbasis $\mathbf{r}_2, \ldots, \mathbf{r}_d$ with real eigenvalues $\lambda_2, \ldots, \lambda_d$. Define $\mathbf{q}_j = \mathbf{U}\mathbf{r}_j$ for $j = 2, \ldots, d$. Then

$$\mathbf{A}\mathbf{q}_j \;=\; \mathbf{AU}\mathbf{r}_j \;=\; \mathbf{UB}\mathbf{r}_j \;=\; \lambda_j \mathbf{U}\mathbf{r}_j \;=\; \lambda_j \mathbf{q}_j.$$

For $i, j \geq 2$, their dot products are

$$\langle \mathbf{q}_i, \mathbf{q}_j \rangle \;=\; \mathbf{r}_i^{\mathsf{T}}\mathbf{U}^{\mathsf{T}}\mathbf{U}\mathbf{r}_j \;=\; \langle \mathbf{r}_i, \mathbf{r}_j \rangle .$$

Thus $\mathbf{q}_2, \ldots, \mathbf{q}_d$ are orthonormal. They are also perpendicular to $\mathbf{q}_1$, because they lie in $\mathrm{Col}(\mathbf{U})$. Hence $\mathbf{q}_1, \ldots, \mathbf{q}_d$ are d orthonormal eigenvectors in $\mathbb{R}^d$. Orthonormal vectors are linearly independent, so they form a basis. All their eigenvalues are real, completing the induction. Together with the matrix identities verified in the main text, this proves Theorem 7.